

SARANYA T R

Senior Software Developer & Solution Architect

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PROFESSIONAL SUMMARY

Senior .NET Integration Engineer and Solution Architect with 9+ years of experience delivering enterprise software for UAE and GCC clients. Completed production integrations between Suprema BioStar 2, OTIS elevator systems, and OTIS card readers using real-time UDP/TCP communication, Windows Services, resilient event queues, and SQL Server. Owns solutions end to end across architecture, backend development, device protocols, cloud infrastructure, deployment, diagnostics, and customer-site testing. Additional expertise includes visitor management, PADS4 digital signage, Microsoft Graph, Azure, React, and AI solutions using Azure OpenAI and RAG.

CORE TECHNICAL SKILLS

- **Backend:** C#, .NET 8, ASP.NET Core, MVC, Web API, Entity Framework Core, Windows Services, background workers
- **Integration:** Suprema BioStar 2, OTIS elevator integration, OTIS card readers, UDP/TCP sockets, WebSockets, REST APIs, device protocols, ACK/NAK handling
- **Cloud & Security:** Microsoft Azure, Azure App Services, Functions, VMs, SQL Managed Instance, Key Vault, Entra ID, OAuth 2.0, OpenID Connect, JWT
- **Data:** SQL Server, stored procedures, query tuning, event-log archiving, MongoDB, MySQL
- **Frontend & Platforms:** React.js, JavaScript, HTML5, CSS3, Bootstrap, PADS4, Microsoft Graph, Exchange, WhatsApp Cloud API
- **AI & Automation:** Azure OpenAI, RAG, embeddings, Ollama, Python, PowerShell, Inno Setup, Windows Task Scheduler

PROFESSIONAL EXPERIENCE

FinesseTech IT Solutions LLC | Dubai, UAE

Oct 2024 - Present

Senior Software Developer & Solution Architect

- Completed a production .NET middleware integration connecting Suprema BioStar 2 biometric access control with OTIS elevator controllers over UDP for approximately 2,000 daily users at a major UAE enterprise.
- Completed OTIS card-reader integration within the VertiVoy solution, including credential capture, destination-floor selection, VIP flows, operating-mode handling, heartbeat checks, and ACK/NAK response processing.
- Designed device communication for OTIS command, destination-entry, heartbeat, and mode-control channels; validated packet behavior using Wireshark and on-site tests with elevator and access-control teams.
- Redesigned real-time access events around a bounded queue and fixed worker model with per-device serial processing, preventing race conditions and protecting the service from unbounded concurrent tasks.
- Rebuilt Suprema synchronization as a standalone Windows Service with per-device modes and shared UDP-client ACK handling; expanded packet identifiers from 8-bit to 32-bit to support production scale.
- Diagnosed and resolved a recurring 24/7 service hang caused by WebSocket cancellation and receive-loop behavior, restoring reliable continuous event processing.
- Built event observability and retention features including queue-wait and processing metrics, local/UTC timestamps, automated SQL Server archiving, and an operational monitoring interface.
- Designed and developed Bonvoy, an enterprise visitor management platform covering pre-registration, QR/PIN and walk-in check-in, Emirates ID reader integration, badges, packages, host approvals, SSO, licensing, and dashboards.
- Built a meeting and room-booking platform integrated with PADS4, Microsoft 365 calendars, WhatsApp Cloud API, and automated visitor and valet notifications.
- Implemented RSA-signed seat licensing, Entra ID SSO, Microsoft Graph email/calendar services, SAFR facial recognition, deployment automation, and HTTP/HTTPS configuration tooling.
- Developed Azure OpenAI and RAG-based assistants for document-grounded responses, plus an offline Ollama assistant with Python orchestration and an animated Godot avatar.

Eureka Infinity Computer Trading LLC | Dubai, UAE

Jul 2023 - Jul 2024

Full Stack Developer

- Developed responsive business applications and REST APIs using .NET, JavaScript, and SQL Server.
- Built a custom ERP solution for currency-exchange operations and improved performance through query optimization and code refactoring.

GW Innovations | India

Nov 2019 - Dec 2021

Senior Software Developer

- Designed, developed, and maintained ASP.NET MVC applications for enterprise clients.
- Optimized SQL Server stored procedures and enhanced existing business-critical applications.

Reon Technologies Pvt. Ltd. | India

Aug 2017 - Sep 2018

Senior Software Developer

- Developed ASP.NET MVC applications and maintained enterprise solutions.
- Improved database performance through SQL Server stored-procedure optimization.

SELECTED ENTERPRISE PROJECTS

- **Suprema and OTIS Integration Middleware:** Production middleware for biometric credentials, elevator destination control, device heartbeat and mode management, packet-level communication, audit logs, and resilient event processing.
- **OTIS Card Reader Integration:** Card-reader workflow for credential detection, authorized-floor selection, VIP behavior, destination-entry commands, and controller acknowledgements within VertiVoy.
- **Visitor Management System:** End-to-end visitor, meeting, identity, badge, package, approval, notification, licensing, and administration platform.
- **Meeting Booking and PADS4:** Room availability, structure-change buffers, Microsoft 365 calendar integration, visitor communication, valet messaging, and signage workflows.
- **AI Assistants:** Azure OpenAI RAG assistant for grounded enterprise knowledge and a fully offline Ollama-based conversational assistant.

EDUCATION

Bachelor of Computer Science | Calicut University | 2008 - 2011

LANGUAGES

English (Fluent) | Malayalam (Native) | Hindi (Basic) | Tamil (Basic)